Changes in Plant Color from Nickel

1. Introduction

Previous analysis found that rgb values taken from an image of a plant can help determine plant health over time. Since the experiment we are examining was created to find out how much nickel pennycress plants can withstand, new questions arose. The goal of the experiment is to find a genotype of pennycress that can uptake high amounts of nickel so that the nickel is removed from the soil. Thus a relationship between nickel level in the soil and color indications was explored. If a relationship is present, we may be able to use color indicators to determine nickel uptake of the plant.

2. Background

As the dataset is longitudinal, tracking plant growth over time, observations are highly correlated with one another. Many models require strict assumptions that the observations are independent. The linear mixed effects model (LMM), however, is specifically made for correlated datasets like this one. When multiple observations are clustered within higher-level groups, such as various images for one plant, standard regression models fall short. Researchers are typically forced to either aggregate the data. This discards data points and valuable within-group differences. Otherwise, they run separate, individual regressions that become noisy because they ignore information across the broader sample. LMMs provide a mathematical trade-off between these two extremes. The model structure partitions the dataset using two distinct components. Fixed Effects, which are constant parameters that do not vary. They estimate the overall population mean or the true regression line across the entire dataset. Or Random Effects, which are parameters modeled as random variables that act as specific deviations from the fixed population mean. Including random effects accounts for the internal correlation of clustered observations, ensuring that the model understands which data points belong to the same overarching group. By using both effects simultaneously, LMMs allow researchers to accurately explore and parse whether a predictor behaves differently within a specific group and between different groups.

3. Exploration

To prevent unnecessary extensive modeling, boxplots of each index for each nickel level were created. In Figure 1, it shows that the median NGRDI of the plants did not significantly change between nickel levels. However, there was a difference in variation between the first 4 levels and the last two. Next, looking at the EXG box plots, the variation is more similar between nickel treatments but the median did decrease as the nickel level increased. Finally, there are a significant number of outliers below the whiskers of the boxplots. The large number of outliers will need to be accounted for.

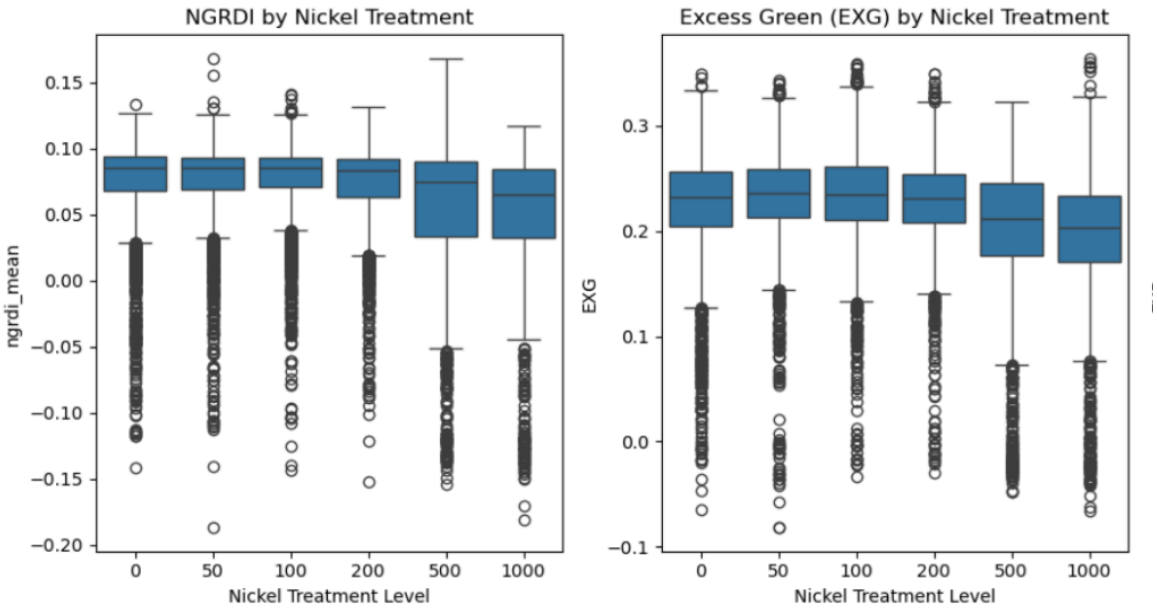


Figure 1: NGRDI and EXG boxplots for nickel treatment levels

The same analysis was done for EXG and VARI as seen in Figure 2. For EXR, the variation increased for the higher nickel levels, with an increasing median. VARI shows similar results to NGRDI with little difference in the median. Also, it has more variation for the high nickel levels. With these results, it was determined that a model should be created using EXR or EXG rather than NGRDI or VARI.

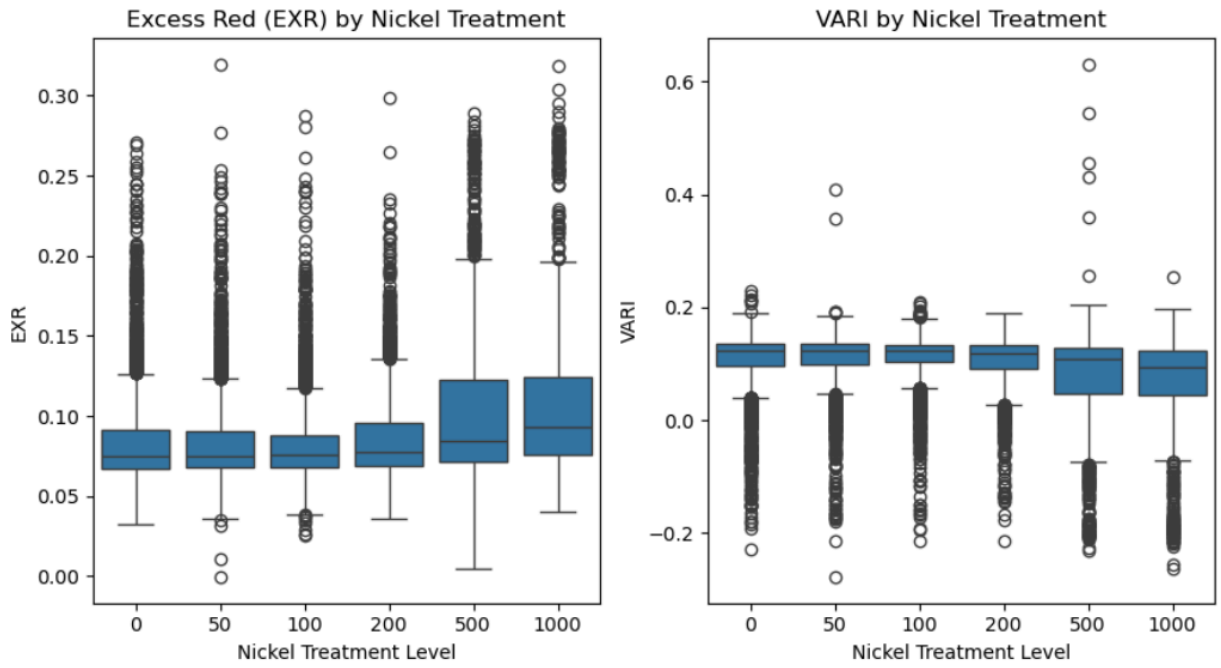


Figure 2: EXR and VARI boxplots for nickel treatment levels

4. Modeling

A linear mixed model was created using EXG and EXR separately. Overall, the models showed similar results but opposite of each other. For example, EXR had a coefficient for the dead plants as a positive relationship and EXG showed a negative relationship. At the baseline of 50 ppm nickel and the time being in the middle of the experiment, the dead plants showed a significantly lower EXG value (and EXR a significantly higher value). This indicates that EXG and EXR do have a relationship with plant death, which confirms what we have seen previously. Additionally, the model showed that the interaction between treatment level and time was significant. Meaning as time progressed, the treatment level changed the colors of the plant when looking at only surviving plants. The same was seen in a three way interaction between treatment, time and the eventual death of the plant. This interaction was highly significant showing the dead plants also saw the same effects from the treatment over time. These results were promising, and seemed to confirm our suspicions on the level of nickel in the plant contributing to the color of the plant. However, after creating the model, a Q-Q plot was made to check the residuals. This Q-Q plot is shown in Figure 3.

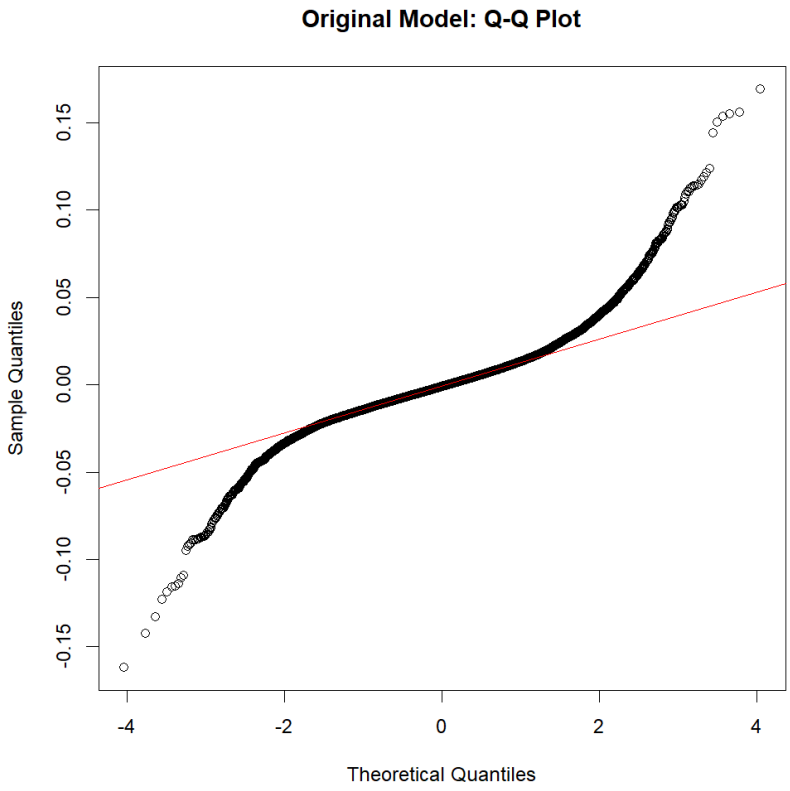


Figure 3: Q-Q plot of residuals from LMM

The graph shows significant tails that veer away from the red normality line. These large tails show that the outliers noted in the boxplots are skewing the results from the LMM. In the future, the model will be tweaked to adjust according to these outliers and new results will be recorded. Unfortunately, due to the non-normal residuals, it is likely the results listed above are inaccurate. The significance of some of the effects were large enough that it is possible the results will hold. Until further tuning is done, it is unclear whether or not the nickel levels change the color values in the rgb imaging of pennycress plants.

5. References

OARC Stats. (2026). *Introduction to linear mixed models*. UCLA. https://stats.oarc.ucla.edu/other/mult-pkg/introduction-to-linear-mixed-models/